AYE
TECH HUB

• INDUSTRIAL ROBOTICS • LESSON 02

Robot Anatomy and Components

Motors · Gearboxes · Controllers · End Effectors · Safety

Lesson 02 of 30 | Industrial Robotics Program

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- Servo motors — the muscle of every industrial robot joint
- Harmonic drives and RV gearboxes — why zero backlash matters
- Encoders: absolute vs incremental position feedback
- The robot controller — architecture, CPU, motion planning
- End effectors: grippers, vacuum, welding torch and tool changers

Robot Mechanical Structure and Servo System

An industrial robot is a precisely engineered mechanical system. Every movement is produced by a **servo motor** → **gearbox** → **link** chain at each joint. The combination of a high-power servo motor and a precision gearbox (harmonic drive or RV reducer) is what gives industrial robots their signature combination of speed, power, and sub-millimetre accuracy.

The Closed-Loop Servo Control System

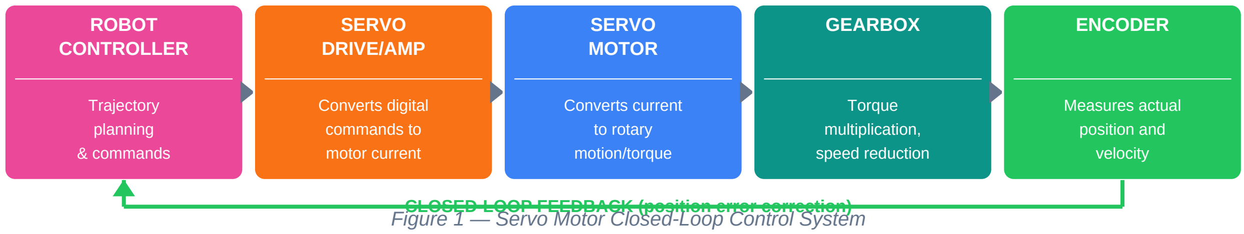

Figure 1 — Servo Motor Closed-Loop Control System

Component	Function	Key Specification
Servo Motor	Converts electrical energy to rotary mechanical motion. Each robot axis has one dedicated servo motor.	Power: 100W–15kW per axis; Speed: 1000–5000 RPM; Torque: matched to gearbox reduction ratio
Servo Drive / Amplifier	Amplifies low-power command signals from controller into high-power current to drive the servo motor.	Accepts ±10V or digital commands; Output current: 1–100A; Includes current and velocity control loops
Gearbox	Reduces high-speed motor rotation to low-speed high-torque joint motion. Critical for precision — zero backlash is essential.	Reduction: 50:1 to 320:1; Backlash: <1 arc-minute (harmonic); Efficiency: 65–99%
Encoder	Measures the actual position and speed of the motor shaft. Sends feedback to controller for closed-loop correction.	Resolution: 17–24 bit (131K–16M counts/rev); Type: Absolute (retains position on power-off) or Incremental
Robot Controller	The brain. Plans trajectories, runs the control loop at 1–4 kHz, manages I/O, runs user program, handles safety.	CPU: Proprietary RISC or industrial PC; Loop rate: 1–4 ms; I/O: Digital, analogue, fieldbus (PROFINET, EtherCAT)

Why zero backlash matters: Backlash is the tiny gap between gear teeth that causes lost motion when direction reverses. In a 6-axis robot, even 0.1° of backlash per joint compounds to significant TCP (Tool Centre Point) position error. Harmonic drives and RV reducers achieve virtually zero backlash which is why robots can repeatedly position to ±0.02 mm.

Gearboxes, Encoders and the Controller

The gearbox and encoder together are the key to robot precision. The gearbox multiplies torque while the encoder verifies exact position. Understanding which gearbox type is used where — and why — is essential knowledge for robot maintenance and specification.

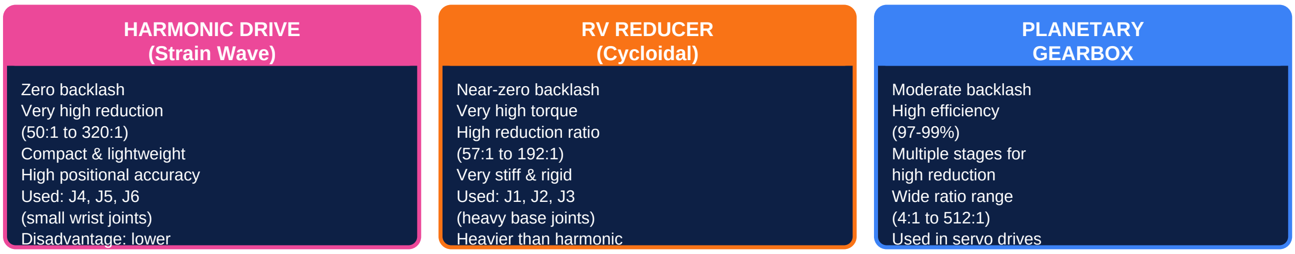

Figure 2 — Robot Gearbox Types: Harmonic Drive / RV Reducer / Planetary

Encoder Type	How It Works	Advantage	Robot Use
Absolute Single-Turn Encoder	Gives unique binary code for every position within one revolution (0–360°). Power-off position is retained.	No homing required on power-up. Knows exact position immediately.	Most modern ABB, KUKA, Fanuc robots — each axis
Absolute Multi-Turn Encoder	Tracks position across multiple revolutions using battery-backed memory or gear train counting mechanism.	Tracks full robot range without homing cycle. Essential for long-stroke axes.	Axes with multi-revolution range; gantry robots
Incremental Encoder	Generates pulses as the shaft rotates. Position is counted from a reference (home) point.	Cheaper, simpler. Unlimited resolution by counting pulses.	Older robot systems; axes where homing is acceptable
Resolver	Analogue device that generates sine/cosine signals proportional to shaft angle. Highly robust.	Immune to electrical noise; works at high temperature; very long life (no electronics).	Harsh environments: foundry robots, welding, high-heat applications

The Robot Controller and Teach Pendant

Controller Component	Function	Key Detail
Main CPU / Processor	Runs the robot program, motion planning, path interpolation	Executes trajectory calculations every 1–4 ms at hardware level
Axis Control Boards	One per robot axis — runs the individual servo control loop	High-speed DSP; communicates with servo drives via EtherCAT/SERCOS
I/O Module	Handles digital and analogue signals to/from PLC, sensors, grippers	Typically 16–64 digital I/O + 4–8 analogue I/O standard
Safety Controller	Monitors speed, force, workspace limits; triggers e-stop if violated	Certified SIL 2 / PLd for collaborative and safety-rated motion
Teach Pendant (TP)	Handheld device for jogging robot, creating programs, I/O monitoring	Touch screen + physical keys + enabling button (dead-man switch)
Power Supply Unit	Converts mains AC to multiple DC voltages for control and drives	24VDC control; 48–400VDC servo bus depending on motor rating

The enabling device (dead-man switch): The teach pendant has a 3-position enabling switch on the back. Fully released = STOP. Half-pressed = robot can move (programming mode). Fully pressed = STOP (panic

grip activates emergency stop). This design means if the operator faints or panics, the robot stops.

End-of-Arm Tooling (EOAT) and Robot Safety

The **end effector** (also called EOAT — End-of-Arm Tooling) is what the robot actually does with the objects it handles. The robot arm delivers position and force; the end effector performs the task. Selecting the right end effector is often more critical than selecting the robot itself.

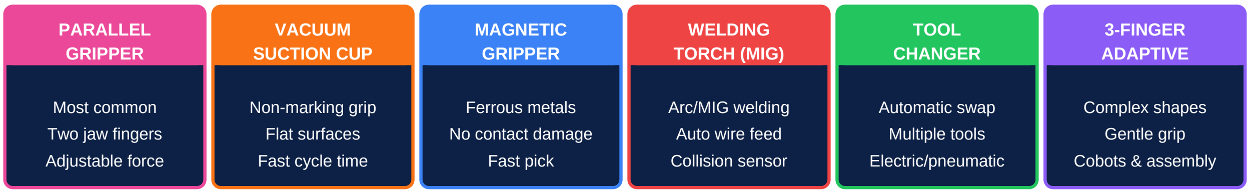

Figure 3 — Common End-of-Arm Tooling (EOAT) Types

EOAT Type	How It Grips	Payload Range	Best Application
Parallel Jaw Gripper (pneumatic/electric)	Two jaws move toward each other symmetrically	1 g – 50 kg	Machined parts, bottles, boxes — most versatile
Vacuum / Suction Cup	Negative pressure (venturi or pump) creates suction on surface	50 g – 100 kg	Flat/smooth surfaces: sheet metal, glass, cardboard, electronics
Magnetic Gripper	Electromagnet or permanent magnet grips ferrous (steel/iron) parts	0.1 – 500 kg+	Steel stamping, sheet metal, castings — no contact damage
MIG/TIG Welding Torch	Delivers wire + shielding gas + electric arc to the weld zone	N/A — fixed tool	Arc welding — automotive body, structural steel, pipe welding
Automatic Tool Changer	Robot docks with tool station to swap EOAT automatically mid-program	Depends on tool	Multi-process cells: drill + measure + pick in same cycle
3-Finger Adaptive Gripper (e.g. Robotiq)	Three independently-actuated fingers conform to object shape	50 g – 10 kg	Mixed parts, irregular shapes, cobots, collaborative assembly

Robot Safety Systems

Safety Device	Function	ISO/IEC Standard
Safety Fencing / Guarding	Physical barrier preventing human entry to robot working envelope during operation	ISO 10218-2: minimum 1.8m height; designed to stop forced entry
Light Curtains / Safety Scanners	Infrared beams detect intrusion. Robot stops immediately if beam is broken.	IEC 61496 (Type 4 for robot applications); response < 20 ms
Safety Mats / Pressure Strips	Floor-level sensors that stop the robot when weight is detected on the mat	EN 1760-1: certified safety mat systems
Emergency Stop (E-Stop)	Mushroom-head pushbutton that immediately cuts servo power. Cannot be bypassed.	ISO 13850: must be red on yellow; self-latching; manual reset required
Speed and Force Limiting (Collaborative Mode)	Robot monitors TCP speed (<250 mm/s) and contact force (<150N) in cobot mode	ISO/TS 15066: Power and Force Limiting (PFL) collaborative operation

Safety-Rated Soft Axes (Safety PLC)	Software-defined workspace limits monitored by certified safety controller	IEC 62061 (SIL 2) / ISO 13849 (PLd): safety function integrity levels

What comes next — ROBOT-003: Robot Kinematics and Coordinate Systems — forward and inverse kinematics explained, the five robot coordinate systems (World, Base, Tool, Wrist, User), joint space vs Cartesian space, and how the robot controller calculates joint angles from TCP positions.